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Drug treatment alters performance in a neural microphysiological system of information processing



Bradley Watmuff¹, Forough Habibollahi¹, Candice Desouza¹, Moein Khajehnejad², Alon Loeffler¹, Koby Baranes^{3,4}, Noah Poulin^{3,4}, Mark Kotter^{3,4,5} & Brett J. Kagan^{1,6} ✉

Assessment of pharmacological intervention on in vitro neural systems often emphasizes molecular and structural changes. However, neural systems fundamentally process and act on information. For preclinical assays to predict drug efficacy, they must model these physiological functions. *DishBrain*, an in vitro synthetic biological intelligence (SBI) assay embodying a neural system in a simulated game-world, enables the quantification of this information-processing capacity, however the question remains whether such a system permits classical pharmacological interrogation and dose-response profiling. Hyperactive glutamatergic dysregulation is linked to neurological disorders including epilepsy, and inducible overexpression of neurogenin 2 (NGN2) in human induced pluripotent stem cells (hiPSCs) generates glutamatergic cultures with dysregulated hyperactivity. We therefore tested three anti-seizure medications (ASMs), phenytoin, perampanel, and carbamazepine, on NGN2 neurons from day 21 of differentiation in this system. The key finding was that, while all compounds altered spontaneous firing, carbamazepine 200 μ M significantly improved gameplay metrics. This marks the first demonstration of altered SBI following exogenous drug treatment. Notably, only inhibitory compounds enhanced goal-directed activity, linking glutamatergic attenuation to performance. Neurocomputational analysis revealed nuanced pharmacological responses during closed-loop stimulation, highlighting insights beyond spontaneous activity metrics.

Predicting the clinical efficacy of a pharmacological intervention targeting a disease with in vitro microphysiological systems (MPS) requires viable assays that physiologically capture the key processes and functions of the tissue or organ it proposes to model. It is likely that no other biological system is as challenging to meaningfully model as neural systems. Neurons are highly dynamic cell types whose primary function is to send, receive and transform electrical signals throughout the brain and periphery. Current MPS models typically provide no stimulation to neural cells, with even state-of-the-art approaches using spontaneous activity or basic evoked activity as a functional outcome¹. Yet, neural systems do not merely fire action potentials, but process structured information and alter both local and population-wide activity in meaningful ways. As a result, in vitro assays which examine the information-processing capabilities of neurons (termed Synthetic Biological Intelligence, or SBI) have recently been described (for a review of SBI and recent advances, see ref. 2). To process information effectively, neural systems must also be able to respond to a variety of

cellular, environmental, and indeed pharmacological inputs with meaningful alterations in their activity^{3–6}. Observing the results of such modulations would be critical for SBI assays to be useful as disease models or as drug discovery screening tools, yet an outstanding question is whether the learning seen in SBI assays is subject to pharmacological intervention.

In this study, we therefore assess for the first time whether function in the recently described SBI assay known as *DishBrain*⁷ can be modulated by the use of pharmacological compounds. We propose that an SBI assay will provide the opportunity to assess not only the functional properties of a disease model, but also nuances that arise in response to pharmacological intervention. By creating a structured information landscape through a real-time closed-loop patterning of electrophysiological stimulation and recording, in vitro neural cultures can be embodied within a simulated game-world environment, in this case representing a simplified version of the game Pong (see the Supplementary Methods and Supplementary Fig. 1 for details of the *DishBrain* closed-loop system, stimulation and feedback

¹Cortical Labs Pte. Ltd., Melbourne, VIC, Australia. ²Turner Institute for Brain and Mental Health, Monash University, Clayton, VIC, Australia. ³Department of Clinical Neurosciences, University of Cambridge, Cambridge, UK. ⁴Wellcome-MRC Cambridge Stem Cell Institute, University of Cambridge, Cambridge, UK. ⁵Bit Bio LTD, Cambridge, UK. ⁶Department of Biochemistry and Pharmacology, University of Melbourne, Parkville, VIC, Australia. ✉e-mail: brett@corticallabs.com

schemes). The observation of related dynamic effects has been replicated by another group, which supports the idea that exploring information processing in *in vitro* neurons is a viable direction for future research⁸. Previous work also demonstrated that cultures of cortical neurons within the *Dish-Brain* system displayed activity changes consistent with goal-directed learning, with significant increases in gameplay performance over time⁷. Moreover, these networks displayed population network activity metrics such as close to critical dynamics⁹. However, prior work has focused on *in vitro* neural cultures intended to model phenotypically “healthy” neural activity. Exploring the behavior of neural cell cultures designed to model disease characteristics would determine the viability of combining an MPS approach with SBI assays to test disease models and screen pharmacological interventions^{10,11}.

Glutamatergic hyperactivity dysregulation is a state where excitatory forebrain neurons’ activity is high frequency, continuous and non-synchronized¹². Post-mortem studies have associated irregular glutamate signaling with a number of neurological and psychiatric conditions¹². A disease commonly associated with glutamatergic hyperactivity is epilepsy^{13,14}. Despite several disease-modifying therapies being available, ~30% of patients with epilepsy do not respond to these treatments¹⁵. As with many other neurological diseases, preclinical research to develop new anti-seizure medications (ASMs) has predominantly leveraged rodent models^{16,17}. Limitations associated with the use of these models have been identified and may, in conjunction with the complexity of the underlying condition and associated pathobiology, contribute to poor clinical trial outcomes¹⁸. At the same time, *in vitro* epileptiform cellular models using human neurons have been developed¹³. The ability to demonstrate compelling dose-response relationships in these *in vitro* models would be highly useful for drug discovery. Moreover, this could enable a personalized medicine approach: culturing cells from patients who display drug-resistant phenotypes; it would also reduce the need for animal testing.

Previous work has leveraged neurogenin 2 (NGN2)-reprogrammed neurons (cultured on an astrocyte layer) to create an *in vitro* model of epilepsy^{19,20}. By utilizing a robust forward programming platform for Optimized inducible Overexpression (OPTi-OX) of NGN2, human induced pluripotent stem cells (hiPSCs) can reproducibly generate neuronal populations with glutamatergic hyperexcitability^{21,22}. However, the extent to which these glutamatergic hyperactive cultures show information-processing changes and whether these cultures may exhibit changes in response to ASMs remains unknown. Here, we aim to use these neuronal cultures that demonstrate glutamatergic hyperexcitability and explore the validity of the system as a simple *in vitro* model of epilepsy. By assessing the electrophysiological development of these cultures over time, including response to specific ASMs, we evaluated the suitability and utility of the *DishBrain* system for exploring disease modeling and drug screening. Finally, given the role of neurocomputational metrics such as neural criticality in predicting phenotypically normal behaviors *in vivo*, and the link these metrics have to epilepsy, we also assessed the population dynamics of these cultures during spontaneous activity and during gameplay in the *DishBrain* system^{9,23,24}.

Specifically, we hypothesized that NGN2-reprogrammed iNeuron cultures would demonstrate increased gameplay performance and phenotypically normalized electrophysiological population dynamics following the administration of existing anti-seizure drugs.

Results

NGN2 iNeuron cultures display glutamatergic and neuronal markers

A broad overview of the study methodology, including cell culture, can be seen in Fig. 1a. To confirm the glutamatergic nature of the OPTi-OX NGN2-iNeurons, we began by examining common neuronal markers at the transcriptional and protein levels. Immunocytochemistry of cultures on day 38 (a time when the cells were clearly active) revealed the presence of the neuronal markers MAP2 and TUJ1, the synaptic proteins SYN1 and PSD-95, as well as a high-level of expression of VGLUT2 (SLC17A6),

a marker of glutamatergic neurons (Fig. 1b, with individual channel images shown in Supplementary Fig. 2a, c, d). With respect to VGLUT2, we segmented DAPI and VGLUT2 objects using FIJI imaging software and quantified the respective overlap (Supplementary Fig. 2b). We also confirmed the presence of GFAP, the glial fibrillary protein present in astrocytes (Supplementary Fig. 2d). We did not observe robust immunolabelling from FOXG1, a transcription factor required for ventral telencephalic development (Supplementary Fig. 2c). Single-cell RNA-Seq experiments further confirmed the upregulation of neuronal, synaptic, and glutamatergic genes, as well as the downregulation of pluripotency genes, from day 0 to day 21 of differentiation. SLC17A7 (VGLUT1) and NGN2 expression peaked at day 7 before returning to baseline, and FOXG1 expression did not appear to change during the timepoints measured (Supplementary Fig. 3).

NGN2 iNeuron cultures display electrophysiological hyperactivity, and pharmacological interventions showcase compound-specific responses

Having confirmed the glutamatergic identity of the neurons in culture, we turned our attention to characterizing their basic functional properties. Cultures developed rapidly over the course of two weeks, from day 7 to day 21, and displayed robust burst and network activity in all channels (Fig. 1c, which shows control activity recorded at day 28). We then measured their activity after the application of carbamazepine, phenytoin, and perampanel. Carbamazepine had a marked effect on reducing mean firing rate after 1 h treatment (Fig. 1c), and we observed a dose-response relationship between increasing carbamazepine dose and reduced mean firing rate (Fig. 1d). At 200 μ M, the reduction in firing activity brought about by carbamazepine treatment was significant (Fig. 1e; $p < 0.05$), however phenytoin and perampanel did not significantly affect the firing rate of cultures at either dose tested.

We next investigated the electrophysiological activity of NGN2 cultures during gameplay, and analyzed the mean firing rate calculated per second, the average variance of the extracted firing rates and the average interspike intervals (ISI) among recorded channels on every culture. The distribution of these metrics is depicted in Fig. 2a, c, e, showcasing comparisons between gameplay and rest conditions under control, as well as across various pharmacological interventions. The plots show an estimated probability density function using Kernel Density Estimation (KDE) with Gaussian smoothing. The negative values in the KDE plots arise from Gaussian smoothing, where the density estimate extends beyond the zero boundary due to data near zero. Statistical comparisons are shown in Fig. 2b, d, f. Additionally, we explored the same electrophysiological metrics across different doses and compared these to the control group as seen in Supplementary Fig. 4a–c. iNeuron cultures also displayed regular, synchronized network activity (Supplementary Fig. 5) at rest and after treatment with the pro-convulsant 4-aminopyridine (4-AP, 100 μ M).

Drawing inspiration from the burst classification methods outlined in Wagenaar et al.²⁵, we extracted quantitative details from the bursting patterns of the recordings from each *in vitro* culture. In order to determine whether the glutamatergic hyperactivity in our cultures was linked to synchronized burst activity, we examined the amount of time each ASM treatment group was in a burst state (Fig. 2g, h). Significant changes in small, medium, and large burst states, respectively, are shown in Fig. 2i–k.

Analysis of gameplay performance under control and pharmacological intervention conditions

To evaluate the game performance of cultures exposed to various pharmacological compounds, three distinct metrics were employed: Average Rally Length (representing the average number of accurate hits per rally), Hit/Miss Ratio (indicating the ratio of accurate hits to missed balls), and Aces to Game Ratio (illustrating the proportion of games where the ball is immediately missed after the initial serve, known as an ace). The estimated probability density functions of these game performance metrics using KDE are visualized across all groups. Figure 3a, c, e, and a statistical significance

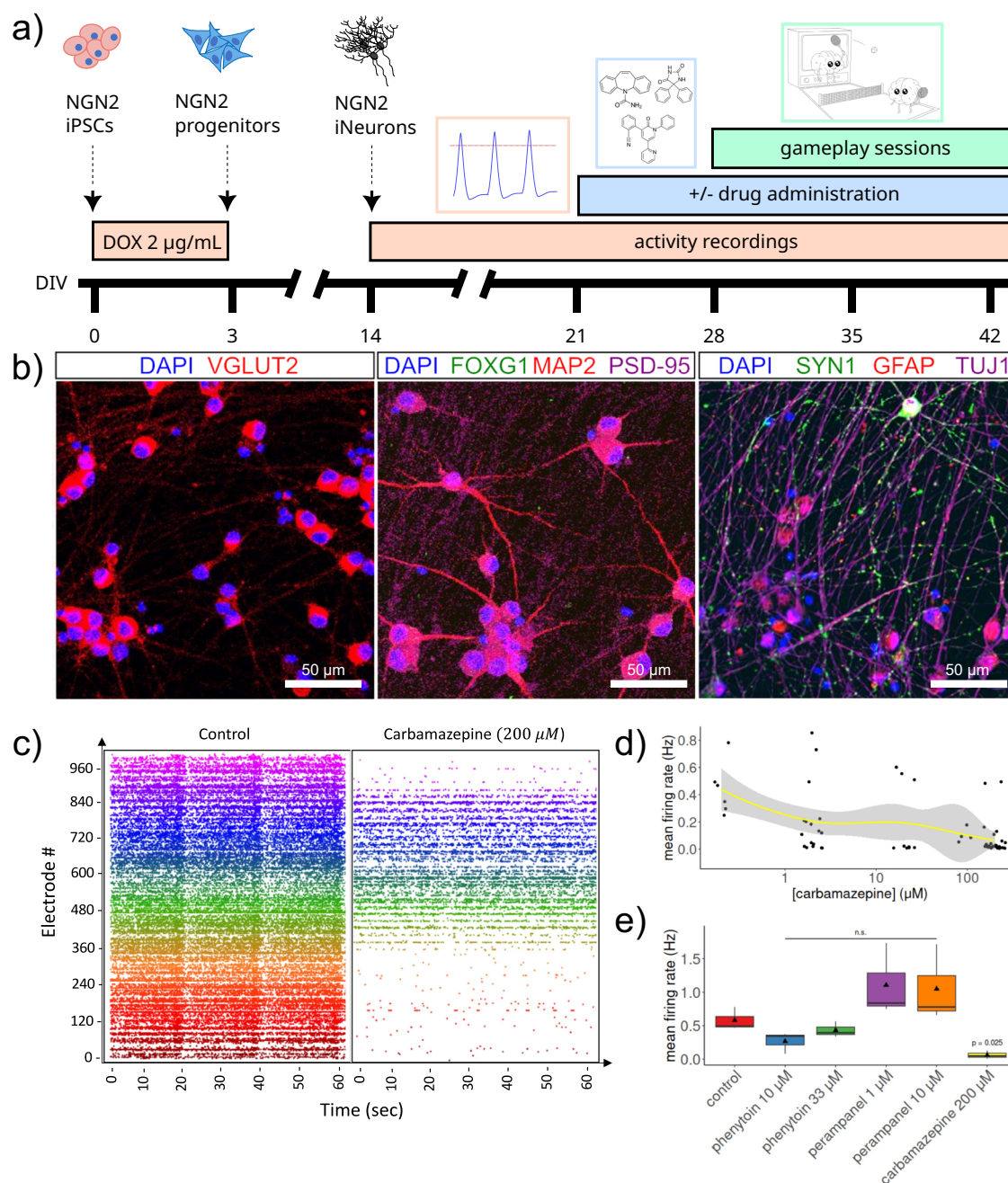


Fig. 1 | Overview of study design and basic NGN2 culture characterization.

a Study overview. **b** Immunocytochemistry of NGN2 cultures at day 38 showing brightfield, DAPI (blue), and VGLUT2 (SLC17A6; red) labeling. Immunocytochemistry in NGN2 cultures showing brightfield, DAPI (blue), FOXG1 (green), MAP2 (red), and PSD-95 (DLG4; magenta) labeling. Immunocytochemistry of NGN2 cultures at day 38 showing DAPI (blue), SYN1 (green), GFAP (red), and TUJ1 (magenta) labeling. Scale bar, 50 μm. **c** Raster plot showing culture firing activity prior to and after carbamazepine (200 μM) administration at day 28. Each

row represents one electrode (of 1024). Each point in the plot represents one detected spike. **d** Relationship between mean firing rates and escalating doses of carbamazepine. Firing rates were calculated per 100 msec for this analysis. **e** Analysis of mean firing rate with administration of different pharmacological compounds. Each group was analyzed by one-way ANOVA with post-hoc Bonferroni test compared to control. Box plots show interquartile range, with bars demonstrating 1.5× interquartile range, the line marks the median, and the black triangle marks the mean.

table are presented for comparing the mean values of different groups (Fig. 3b, d, f). Subsequently, the same metrics are computed for various doses of each administrated drug (Supplementary Fig. 6a–c).

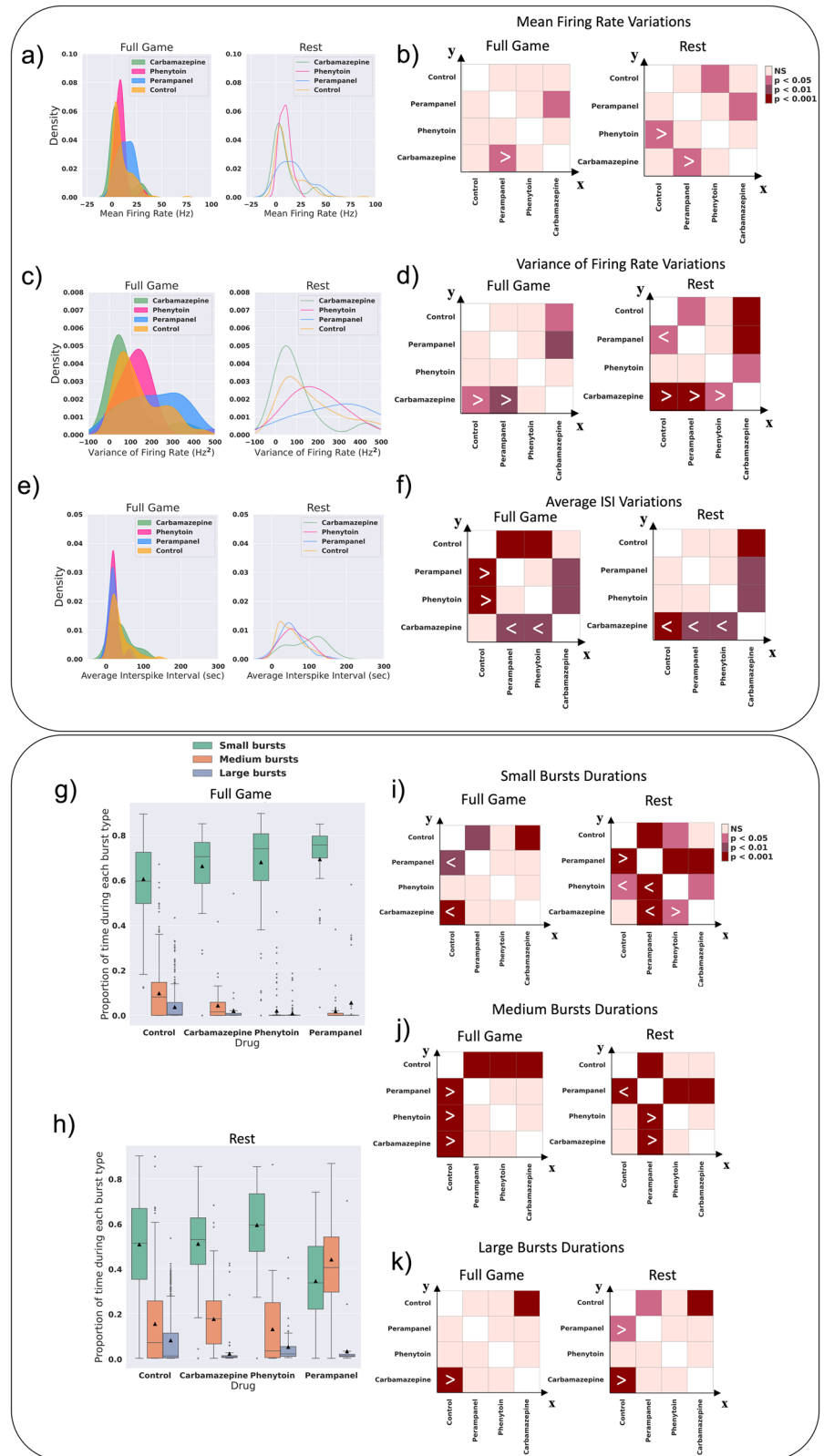
We next examined the effect of the duration of gameplay on our analysis. Figure 3g–i represents the percentage of changes in average rally length compared to rest under control and all the pharmacological interventions when studying the first and second half of each 15-min-long recording. The statistical significance table is presented for comparing the mean values of different groups in Fig. 3j–l. The high dose of carbamazepine

(200 μM) was the only group showing a significant improvement in time in terms of the average rally length compared to rest.

Bursting patterns of activity under control and pharmacological intervention conditions

We further explored bursting patterns of activity under both control and pharmacological intervention conditions as described above. In certain recordings, bursts displayed highly consistent sizes, while others exhibited a broad range of burst sizes.

Fig. 2 | Neural firing and burst dynamics under control and pharmacological intervention conditions. a, c, e Estimated probability density function plots using KDE with Gaussian smoothing to compare Mean Firing Rate, Variance of Firing Rate, and Average Interspike Interval (ISI) in both rest and full game sessions. The negative values in the KDE plots arise from Gaussian smoothing, where the density estimate extends beyond the zero boundary due to data near zero. The sample sizes (number of recording sessions) are $n = 42$, $n = 42$ (Carbamazepine: rest, full game), $n = 24$, $n = 24$ (Phenytoin: rest, full game), $n = 24$, $n = 23$ (Perampanel: rest, full game), $n = 354$, $n = 400$ (Control: rest, full game). **b, d, f** Post-hoc Games-Howell test between all the rest and all the full game groups. $>x > y$ by significance level and $<x < y$ by significance level. **g, h** Boxplots illustrating the average total proportion of time spent during the occurrence of each burst type (small, medium, and large) under control and pharmacological intervention conditions. sessions. The sample sizes are $n = 354$, $n = 400$ (Control: rest, full game), $n = 42$, $n = 42$ (Carbamazepine: rest, full game), $n = 24$, $n = 24$ (Phenytoin: rest, full game), and $n = 24$, $n = 23$ (Perampanel: rest, full game). Boxplots show interquartile range, with bars demonstrating $1.5\times$ interquartile range, the line marks the median, and the black triangle marks the mean. Error bands = 1 SE. **i–k** The significance heatmaps depict the post-hoc Games-Howell test results comparing control and different pharmacological intervention conditions during rest and the full game. On the axes, “ $>$ ” indicates that the group on the x -axis is greater than the group on the y -axis at the given significance level, and “ $<$ ” indicates that $x < y$ at the given significance level.

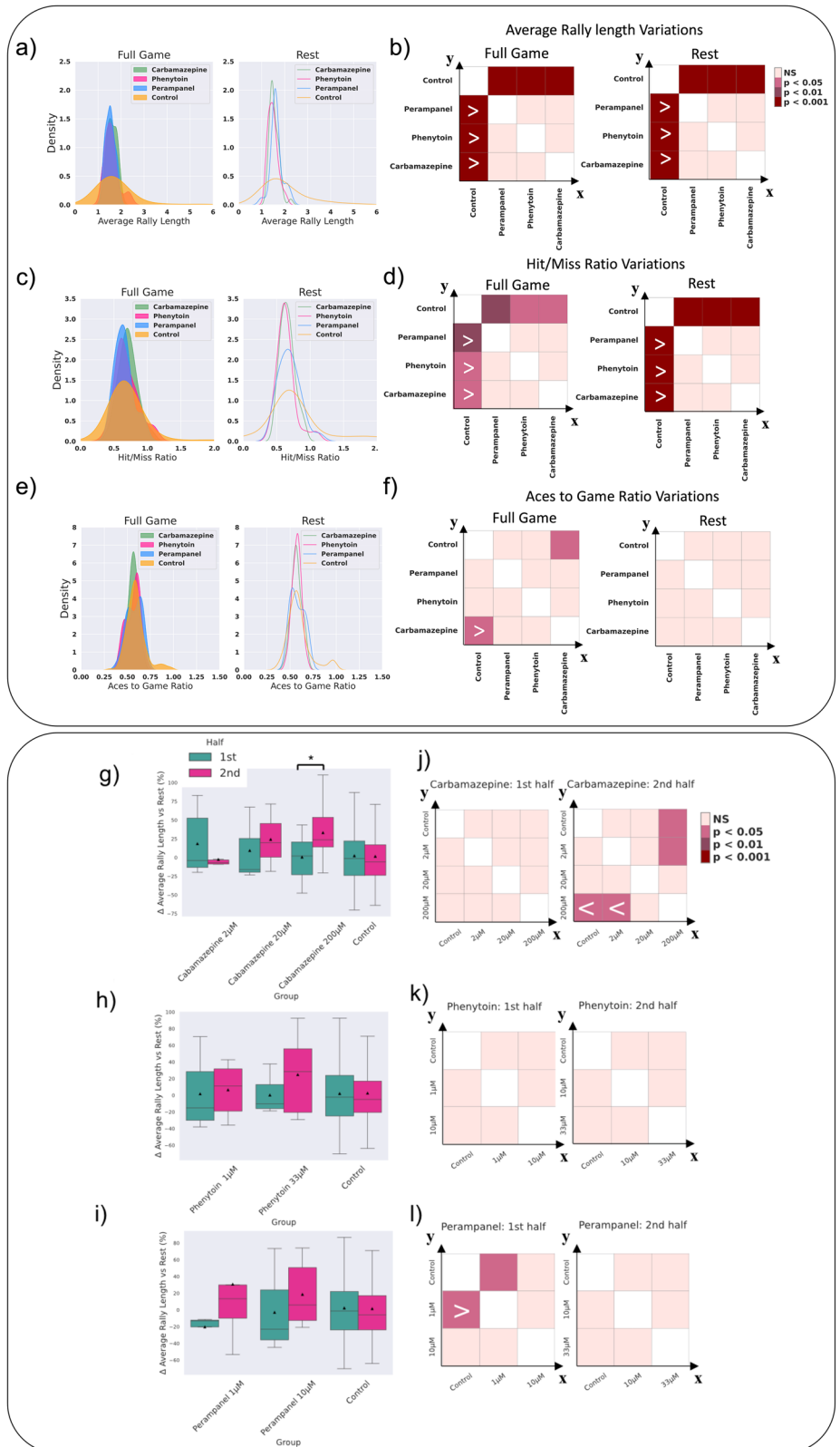


We proceeded to tally the number of small, medium, and large bursts in all recordings. Furthermore, we normalized the counts of various burst sizes in all subsequent sessions to the count of those burst sizes in the initial rest session recording. The average duration of burst activity in recordings at rest and during gameplay is shown in Supplementary Fig. 7. Figure 4a–c shows the number of extracted small, medium, and large bursts from full game and rest sessions under control and all the pharmacological interventions.

Post-hoc statistical significance for the comparisons between groups is visually represented in Fig. 4d–f. The same metrics are then illustrated for various administered doses of each drug compared to the control group in Fig. 4g–o. The count of small bursts dropped significantly both during rest and gameplay with the administration of all compounds. The count of large bursts also significantly decreased in both carbamazepine and phenytoin groups during gameplay relative to control and significantly decreased in the

Fig. 3 | Game performance analysis under control and pharmacological intervention conditions.

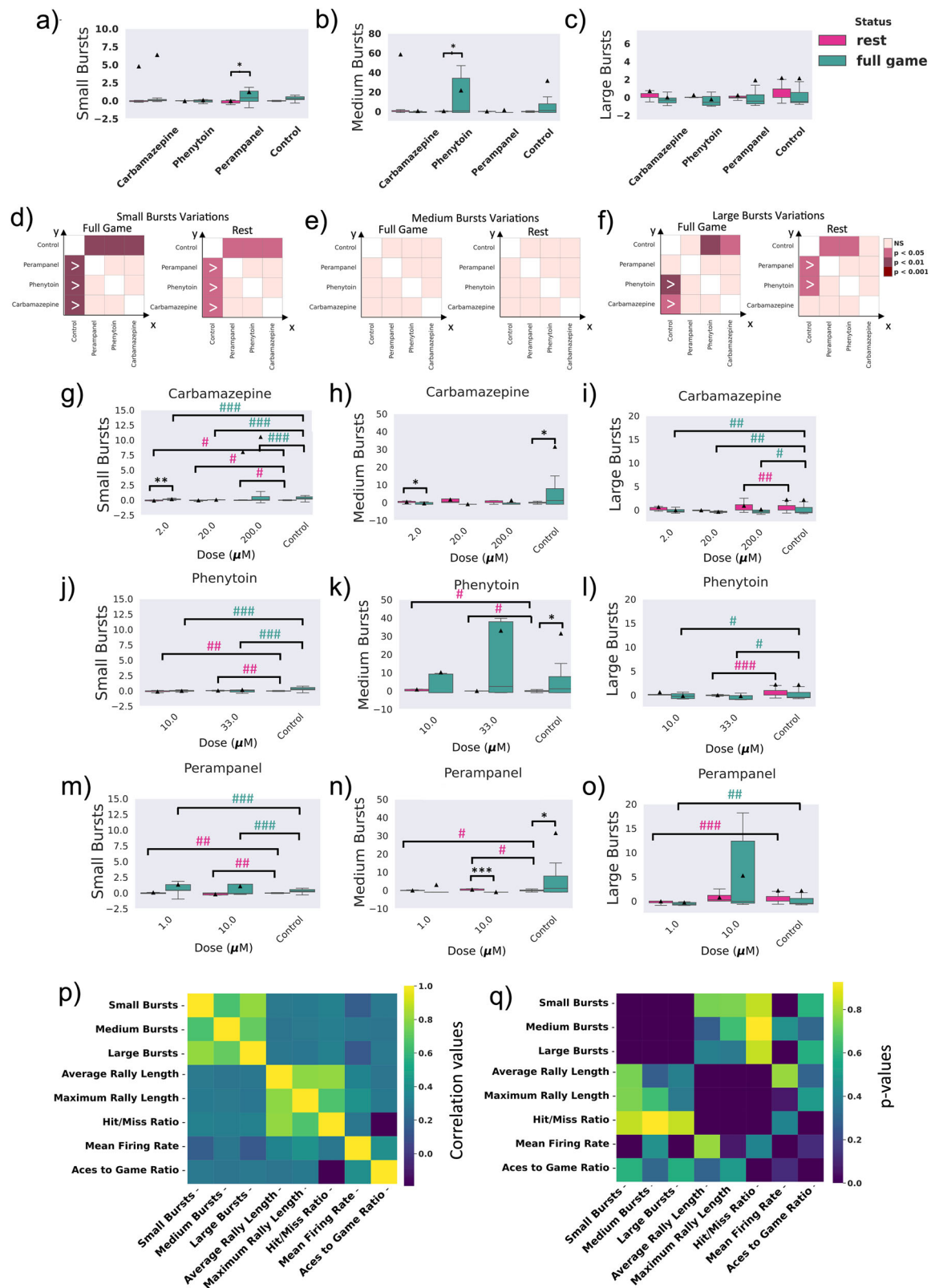
a, c, e Comparison of Average Rally Length, Hit/Miss Ratio, and Aces to Game Ratio in both rest and full game sessions. Kernel Density Estimation plots are used, applying Gaussian smoothing to estimate the probability density function of the data. The sample sizes are $n = 42$, $n = 42$ (Carbamazepine: rest, full game), $n = 24$, $n = 24$ (Phenytoin: rest, full game), $n = 24$, $n = 23$ (Perampanel: rest, full game), $n = 354$, $n = 400$ (Control: rest, full game). **b, d, f** The significance heatmaps depict the post-hoc Games-Howell test results comparing control and different pharmacological intervention conditions during rest and the full game. On the axes, “>” indicates that the group on the x -axis is greater than the group on the y -axis at the given significance level, and “<” indicates that $x < y$ at the given significance level. **g, h, i** Comparison of Average Rally Length changes in full game sessions vs rest when comparing the first and second half of each recording. * indicates $p < 0.05$. Box plots show interquartile range, with bars demonstrating 1.5 $\times$ interquartile range, the line marks the median, and the black triangle marks the mean. Error bands = 1 SE. **j, k, l** The significance heatmaps depict the post-hoc Games-Howell test results comparing control as well as various doses of each pharmacological intervention condition during the first and second half of full game sessions. On the axes, “>” indicates that the group on the x -axis is greater than the group on the y -axis at the given significance level, and “<” indicates that $x < y$ at the given significance level.



phenytoin and perampanel-treated groups during rest compared to the control group. Figure 4p shows the Pearson correlation values between each of the burst pattern metrics and different electrophysiological and game performance metrics, including the metrics introduced previously. The corresponding p values are illustrated in Fig. 4q. Interestingly, the Mean Firing Rate appears to be negatively and significantly correlated with the number of small and large bursts in the cultures.

Critical dynamics of cultures under control and pharmacological intervention conditions

Dynamical systems undergo transitions between ordered and disordered states, and the concept of “criticality” emerges when the system resides at the boundary between these states—where the input is neither strongly damped nor excessively amplified²⁶. Failures in adaptive criticality may contribute to impairments in brain function, such as those observed in dementia or



epilepsy²⁷. Building on insights from recent literature^{9,28}, an avalanche analysis was conducted to examine the network in relation to its proximity to criticality.

The start and end of an avalanche are determined by crossing a threshold of network activity, with spikes from any or all neurons within a specified region capable of triggering an avalanche. The scale-free dynamics

of detected neuronal avalanches, along with metrics such as the Deviation from Criticality Coefficient (DCC), Branching Ratio (BR), and Shape Collapse error (SC error), were assessed to gauge whether the recordings were poised near criticality⁹. A lower DCC and SC error, along with a BR closer to 1, serve as indicators of a state near criticality. Figure 5a–c show the number of extracted DCC, BR, and SC errors from full game and rest sessions under

Fig. 4 | Burst pattern analysis under control and pharmacological intervention conditions. **a–c** The normalized number of small, medium, and large bursts for rest and full game sessions of all control and drug-treated cultures. The first rest session recording of each experiment is used for normalization. The sample sizes are $n = 35$, $n = 35$ (Carbamazepine: rest, full game), $n = 20$, $n = 20$ (Phenytoin: rest, full game), $n = 20$, $n = 20$ (Perampanel: rest, full game), $n = 97$, $n = 95$ (Control: rest, full game) from left to right. One-way ANOVA test. **d–f** The significance heatmaps depict the post-hoc Games-Howell test results comparing control and different pharmacological intervention conditions during rest and the full game. On the axes, “>” indicates

that $x > y$ at the given significance level, and “<” indicates that $x < y$ at the given significance level. **g–o** Comparison of the same burst pattern metrics between different administered doses of each drug. One-way ANOVA test was conducted within each group, and post-hoc Tukey’s test was conducted between all the rest and all the full game groups. # or * indicate $p < 0.05$, ## or ** indicate $p < 0.01$, ### or *** indicate $p < 0.001$. Boxplots show interquartile range, with bars demonstrating $1.5 \times$ interquartile range, the line marks the median and the black triangle marks the mean. Error bands = 1 SE. **p, q** Pearson correlation test between the burst pattern and the game performance metrics.

control and all the pharmacological interventions. Post-hoc statistical significance for the comparisons between groups is visually represented in Fig. 5d–f. The same metrics are then illustrated for various administered doses of each drug compared to the control group in Fig. 5g–o. The results demonstrate that criticality metrics are greatly disrupted both under control and pharmacological intervention in both game states. Some small variations are observed, such as decreased DCC and increased BR values in low doses of carbamazepine (2 μM) or decreased SC error in higher doses (200 μM) compared to control during gameplay. The BR also displays a significant increase when comparing gameplay to rest state in 200 μM carbamazepine. Overall, there is limited expression of the various criticality metrics in these cultures whose neurons are nearly all glutamatergic. This is supported by previous literature, which showed that critical state dynamics arise from a balance of excitatory and inhibitory neuronal networks^{26,29}.

Figure 5p shows the Pearson correlation values between each of the criticality metrics and different electrophysiological and game performance metrics, including the metrics introduced previously. The corresponding p values are illustrated in Fig. 5q. Culture game performance measured in Hit/Miss Ratio is negatively and significantly correlated with SC error, while Mean Firing Rate appears to have a positive and significant correlation with this criticality metric.

Functional connectivity of cultures under control and pharmacological intervention conditions

In order to determine how ASM administration affected underlying network activity at rest and during gameplay, we used neuronal spiking activity to construct functional connectivity networks for all recordings. Due to the extensive duration of the recordings at a 20 kHz sampling frequency, the resulting time series data was substantial. Following the approach outlined in Khajehnejad³⁰, dimensionality reduction techniques, particularly t-SNE³¹, were employed to enhance computational efficiency and improve data interpretability. The 3-dimensional representations obtained using t-SNE for both rest and gameplay recordings were utilized to investigate latent network structures. To evaluate the effectiveness of these low-dimensional representations, each rest and gameplay recording session was first divided into two equal time intervals prior to applying dimensionality reduction. Figure 6a, c, e, g shows the results after color-labeling the first and second halves of the recording sessions for sample cultures under both gameplay and rest conditions. The visualization shows that while the two halves of gameplay sessions are distinguishable using t-SNE, the distinction is less evident during rest sessions. Functional connectivity network matrices were then constructed using zero-lag Pearson correlations for each gameplay or rest session recording, with the 1024 channels as nodes and weighted edges representing functional connectivity. Only edges with Pearson correlation absolute values above 0.7 were retained.

Considering that only a fraction of neurons fire at any given time within complex neural networks, the same method introduced in ref. 30 was utilized to reduce computational complexity while preserving network dynamical properties. This involved identifying a subset of recorded channels that likely monitored neuronal populations specifically attuned to the ongoing task. Tucker decomposition via higher-order orthogonal iteration on tensor data from all lower-dimensional representations of the recordings was employed to identify a consistent subset of channels across all neuronal cultures. K-medoid clustering was then applied to partition the data into 30 clusters, and the corresponding ‘medoids’ were extracted as the

mutual representative channels amongst all cultures. Finally, another network matrix using Pearson correlation was built with these 30 channels as nodes. The resulting functional connectivity networks were derived separately for the first and last 5 minutes of every recording.

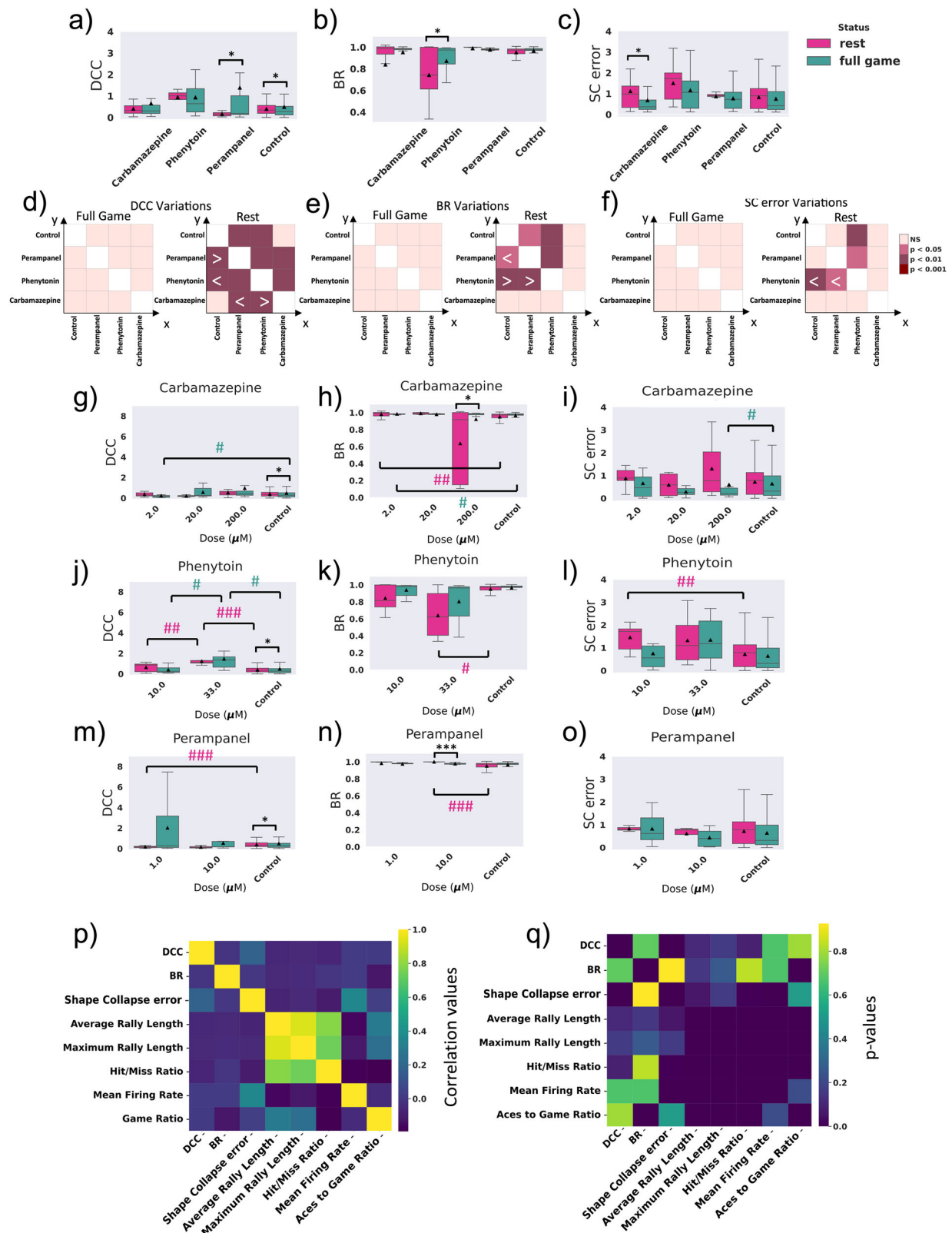
Figure 6b, d, f, h illustrate the heatmap of correlation values (i.e., edge weights) in the constructed functional connectivity networks for the smaller network (30 representative channels) and during rest and gameplay sessions. Heatmaps of the full network (1024 channels) are found in Supplementary Fig. 8.

Next, to evaluate the dynamics of the functional connectivity networks in time, we divided each of the rest of the gameplay sessions into 5-minute intervals and visualized the connectivity networks for the first and last 5 minutes of each group.

As a result of all pharmacological interventions, these functional connectivity networks using the low-dimensional representations of neural activity during rest and gameplay reveal distinct patterns evolving over time, specifically during gameplay. This contrasts with the absence of such patterns during the spontaneous activity observed at rest, as seen in Fig. 6i–p. This difference is evident when considering both the full network (1024 channels; Supplementary Fig. 9) and the smaller network (30 representative channels; Fig. 6i–p).

We then studied various macroscopic metrics of the connectivity network for the control group and all the different doses of the pharmacological interventions. The measured metrics encompassed: 1. Average Weight: This metric illustrates the mean value of Pearson Correlations calculated between pairs of spiking time series in each culture. 2. Modularity Index: This index quantifies the extent to which a network can be partitioned into distinct and internally cohesive groups or communities. It evaluates how nodes form clusters that are more densely connected within themselves than with nodes outside the cluster. 3. Clustering Coefficient: This measure quantifies the tendency of nodes to form clusters or highly connected groups, providing insights into the local connectivity of the network by assessing the likelihood that neighbors of a particular node are connected to each other. Figure 7a–l represents these derived network metrics for the gameplay and rest sessions under pharmacological intervention conditions, as compared to control conditions.

We then studied the changes in the introduced macroscopic network metrics from pharmacological interventions when comparing the first and last 5 minutes of recordings (see Fig. 7m–u). We utilize the first and last 5-minute time intervals for both the full network and the smaller network. Supplementary Fig. 10 represents the evolution of the macroscopic network metrics over time under the various pharmacological interventions using the full network. Supplementary Fig. 11 studies the same network statistics for various doses of administered drugs. Carbamazepine appears to show a significant change in network dynamics only when administered at its high dose (200 μM). Increasing average weight and clustering coefficient, as well as decreasing modularity index, are observed during gameplay but not in rest for all of the administered drugs. While these differences in gameplay are more significant for phenytoin and perampanel, specifically using the full network in Supplementary Fig. 8, they rapidly lose the changes which arise during the initial embodiment in the gameplay environment –such as those observed in the first 5 minutes– and return to baseline levels observed during rest. This returning to baseline effect is not observed after carbamazepine administration.



Discussion

By using an in vitro iPSC neuron model characterized by glutamatergic hyperactivity, we intended to explore the behavior of these cells in a structured information landscape and the impact of pharmacological interventions intended to suppress electrophysiological hyperactivity. It was proposed that modeling dynamic environments through closed-loop

electrophysiological stimulation and recording would provide an enhanced MPS-based method to embody simple neural cultures in a simulated game-world and may offer additional information over existing drug-response assays traditionally used in MPS. The first step in this experiment was to establish a suitable disease model. Using the OPTi-OX system²², NGN2-iNeurons were generated, which displayed elevated levels of glutamatergic

Fig. 5 | Criticality analysis under control and pharmacological intervention conditions. a–c The Deviation from Criticality Coefficient (DCC), Branching Ratio (BR), and Shape Collapse error (SC error) for rest and full game sessions of all control and drug-treated cultures. The sample sizes are $n = 24$, $n = 36$ (Carbamazepine: rest, full game), $n = 20$, $n = 24$ (Phenytoin: rest, full game), $n = 12$, $n = 19$ (Perampanel: rest, full game), $n = 288$, $n = 369$ (Control: rest, full game) from left to right. One-way ANOVA test. **d–f** The significance heatmaps depict the post-hoc Games-Howell test results comparing control and different pharmacological intervention conditions during rest and the full game. On the axes, “>” indicates that $x > y$ at the given

significance level, and “<” indicates that $x < y$ at the given significance level. **g–o** Comparison of the same criticality metrics between different administered doses of each drug. One-way ANOVA test within each group and post-hoc Tukey’s test between all the rest and all the full game groups. # or * indicate $p < 0.05$, ## or ** indicate $p < 0.01$, ### or *** indicate $p < 0.001$. Boxplots show interquartile range, with bars demonstrating $1.5 \times$ interquartile range, the line marks the median and the black triangle marks the mean. Error bands = 1 SE. **p–q** Pearson correlation test between the criticality and the game performance metrics. **p** Shows the correlation values.

markers. These NGN2-iNeurons demonstrated heightened spontaneous spiking, greatly exceeding that of primary cortical rodent cultures and dorsal forebrain cultures intended to model healthy activity⁷. It is also interesting to note that the spiking activity of these OPTi-OX inducible lines far exceeds the activity of virus-generated NGN2 cultures previously tested in the *DishBrain* system, which did show evidence of statistically significant learning over time without pharmacological intervention⁷. These results may reflect a greater consistency of NGN2 overexpression with the OPTi-OX system relative to viral methods, or instead that the OPTi-OX system results in similar transgene expression but with more hyperactivity in the resulting cultures; to conclusively determine this, further work is required. Either way, these data suggest that OPTi-OX NGN2-iNeurons can produce a robust model of glutamatergic hyperactivity^{13,14,19}.

The evidence whether these OPTi-OX NGN2 cultures were suitable as an epilepsy model was more nuanced. The NGN2 cultures displayed increased activity coupled with a limited inhibitory network. Some evidence of recurrent epileptiform activity, with large network bursts comparable to seizures and suppressed upon ASM administration, was also observed, but the complexity of this activity was limited due to the simple networks within these cultures. Yet some further support for this model’s functionality was observed in the compound-specific electrophysiological modulation achieved when various ASMs were introduced to the cultures. Preliminary pharmacological treatments of these cultures showed a reduction in culture activity and changes in bursting patterns after an hour of exposure to ASMs. Interestingly, the magnitude of bursting pattern changes, particularly for large bursts, differed between ASMs. As expected, we observed concentration-response relationships between the dose of drugs administered, and mean culture firing rate, indicating the presence of the drug’s respective receptor targets, and the sensitivity of the glutamatergic hyperexcitable phenotype to pharmacological manipulation. The mechanism of action of all three compounds tested is established: perampanel binds non-competitively and selectively to AMPA-type glutamate receptors³², and both phenytoin and carbamazepine block voltage gated sodium channels on the surface of neuronal cell membranes^{33–35}. The effect of ASM administration is a reduction in aberrant neuronal excitability associated with depolarization and action potentials, leading to a decreased likelihood of seizures. One limitation of our study is that, while the neural systems were supplemented with astrocytes, the model lacks other supporting cell types, including interneurons and an appropriate balance of inhibitory cells, which are likely important to capture more nuances expressed in epilepsy clinically. Despite this limitation, the reduction in firing rate and changes in burst state we measured in this study are consistent with the mechanism of action these compounds display clinically, and led us to question if the acute drug effect would translate to modulation of gameplay performance in the full *DishBrain* gameplay assay.

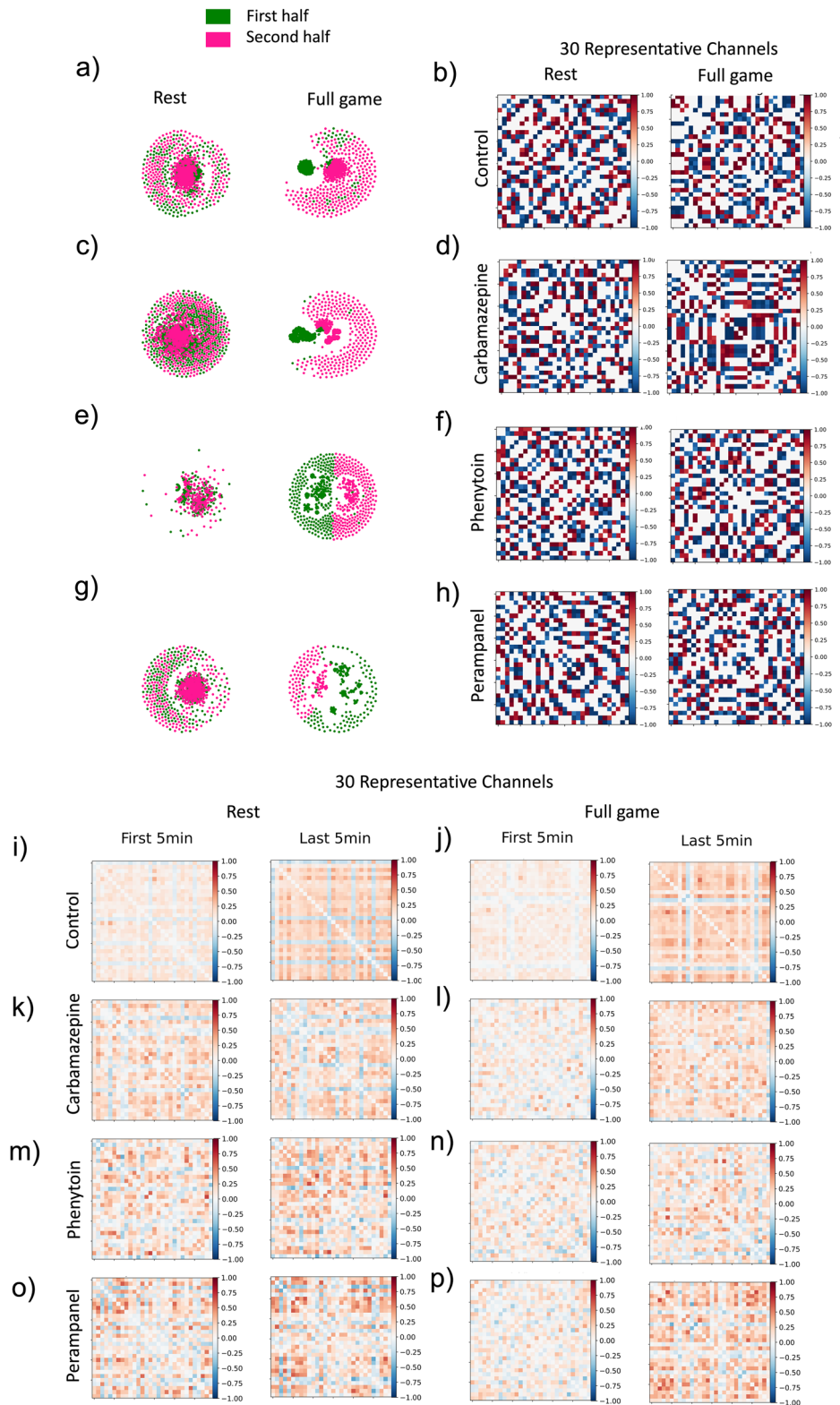
When examining the overall gameplay performance characteristics of the neural cultures, significantly more variance was observed in control cultures without pharmacological intervention. Despite this variation, mean culture performance, after treatment with carbamazepine, was significantly improved (more hits, longer rallies, fewer aces) when compared to control and other treatment groups. Similarly, only the 200 μ M carbamazepine group showed overall significantly higher performance on average rally length in gameplay compared to rest. This result becomes more pronounced when looking at learning over time instead of just between groups overall, consistent with previous work in this area^{7,36}. Despite these statistically

significant differences, the effect sizes are notably smaller than in previous work. Of note, no statistically significant learning effects were observed over time in untreated cultures. Given the substantially different electrophysiological profile of the OPTi-OX NGN2-iNeurons tested in this work relative to the virally induced NGN2 cells tested previously, this may reflect the possibility that hyperactivity is not conducive to information processing and learning. This is supported by the observation that these small but significant learning effects occurred after the loss of the majority of activity in the presence of 200 μ M carbamazepine. If so, this would provide some support for the use of these cells as a disease model, in line with previous work¹⁹.

Moreover, extending beyond whether neural cultures showcased any overt evidence of improved gameplay performance, it is also interesting to compare and contrast different electrophysiological phenomena while cells are embodied in the structured information landscape vs engaged only in spontaneous activity. For example, looking at different sizes of network burst activity, an increased dosage of phenytoin appears to significantly increase medium bursts during gameplay vs controls and the lower dose, but not during rest; perampanel has a similar effect on small bursts. These effects appear reasonably consistent across different dosages and likely reflect the nuances of the various compounds’ actions on neural function both with and without structured stimulation. Without the ability to embody neural cultures in these closed-loop systems, it would not be possible to extract this data, providing further evidence as to the benefit of augmenting neural MPS systems with this approach for future studies. Examining simple pairwise correlations did not reveal a direct and obvious link between these bursting patterns and gameplay characteristics. However, future work focused on explaining these mechanisms could aim to build more nuanced computational models to understand whether any moderated or mediated pathways exist. Similarly, previous work established that when neural systems intended to model healthy activity were placed in a closed-loop structured information landscape, they would show significantly closer to critical dynamics compared to when engaged in spontaneous activity alone⁹. Given that criticality has been identified as a clinically relevant metric and is specifically associated with epileptiform activity, it was a meaningful metric to consider in this work as well^{23,24}. Consistent with this previous literature, criticality was found to be greatly disrupted in these cultures, even though some small evidence of variations due to pharmacological intervention and game state was found. The limited expression of the various criticality metrics in these cultures, whose neurons are nearly all glutamatergic, also supports previous literature, which showed that critical state dynamics arise from a balance of excitatory and inhibitory neuronal networks^{29,37}. Thus, the lack of excitatory-inhibitory balance in this work may explain the limited observations of critical dynamics and learning effects, relative to previously reported levels⁹. Despite this, significant correlations were still observed between some criticality and performance metrics. While this was not replicated across all measures, it provides support for further consideration of critical dynamics as a marker of preclinical pharmacological intervention.

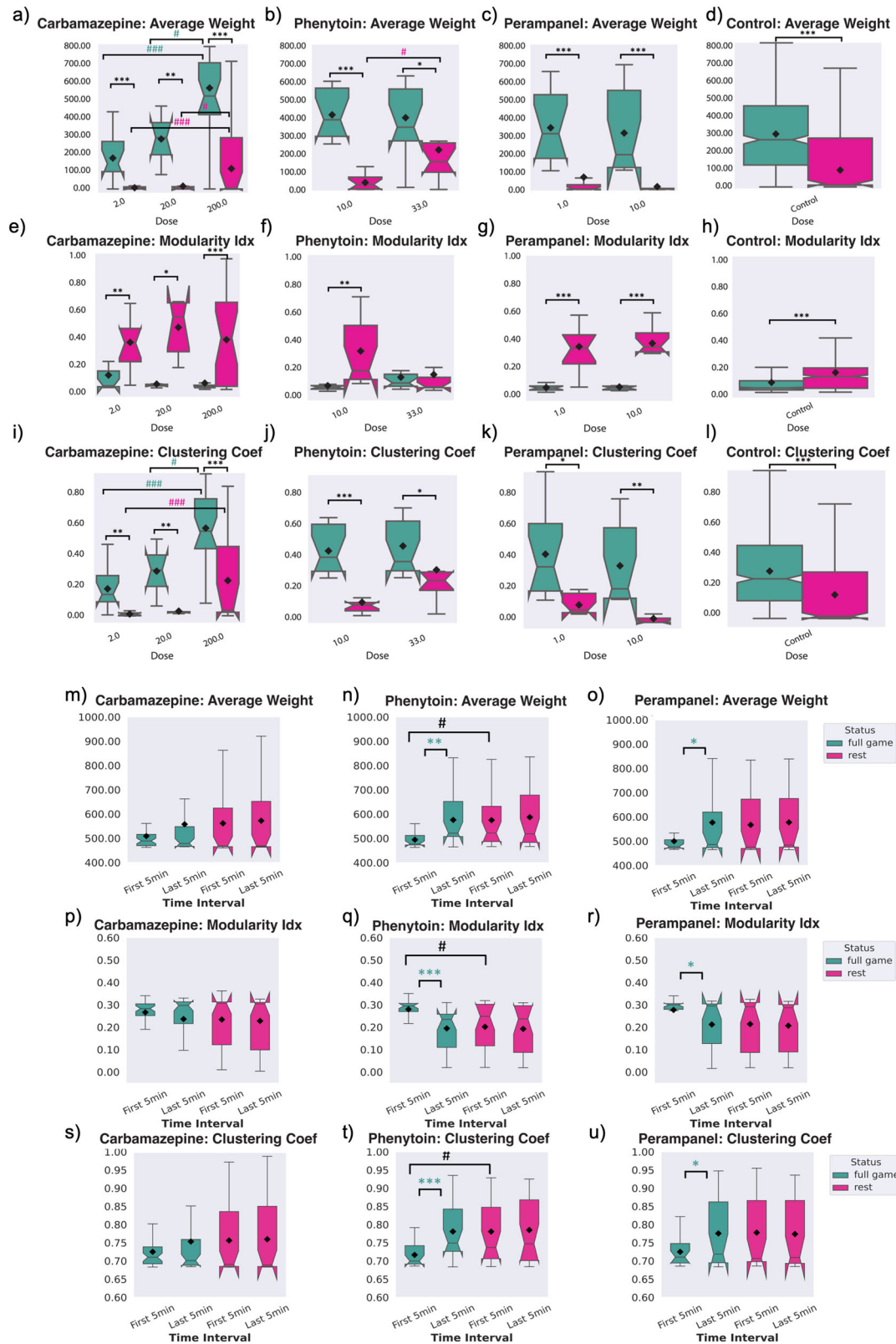
Finally, we also explored the functional connectivity within neural cultures, both unstimulated and when embodied in the gameplay environment. Previous research has found that functional connectivity is linked to key behavioral changes within in vitro cultures when embodied in structured information landscapes^{8,30}. This work is replicated here

Fig. 6 | Functional connectivity networks of rest and full game sessions under control and pharmacological interventions. a, c, e, g Low-dimensional representation of samples of rest and full game sessions using t-SNE under control and pharmacological intervention conditions (Carbamazepine, Phenytoin, and Perampanel, from top to bottom, respectively). The green and purple dots are the channel representations in the embedding space in the first and second halves of the recordings, respectively. **b, d, f, h** Heatmaps illustrating the average edge weights between the 30 representative channels in the Pearson Correlation network constructed from all rest and all full game recordings under control and pharmacological intervention conditions. The sample sizes are $n = 42$, $n = 43$ (Carbamazepine: rest, full game), $n = 24$, $n = 24$ (Phenytoin: rest, full game), $n = 24$, $n = 24$ (Perampanel: rest, full game), $n = 354$, $n = 400$ (Control: rest, full game). One-way ANOVA test. **i–p** Changes in the functional connectivity networks of rest and full game sessions under control and pharmacological interventions. Heatmaps illustrate the change in average edge weights between the 30 representative channels in the Pearson Correlation network constructed from the first and last 5 minutes of all rest and all full game recordings under control and pharmacological intervention conditions.



using low-dimensional representations of neural activity. Clear differences can be observed during gameplay, which are not seen during the purely spontaneous activity displayed during rest. This represents a degree of reorganization in neural cultures, even if such organization does not extend towards robust metrics of improved gameplay performance. When examining the average edge weights, either looking at all channels or the condensed 30 representative channels, it is apparent that there exists far more structure in the changes occurring in the cultures

during the gameplay compared to rest, where minimal structured changes appear. Nonetheless, while looking at the network summary statistics between different pharmacological interventions during gameplay and rest, a more nuanced picture emerges. In particular, it appears that administration of either phenytoin or perampanel results in the average weight, modularity index, and clustering coefficient metrics rapidly losing the changes that arise during the initial embodiment in the gameplay closed-loop environment, and returning to baseline levels



observed during rest. In contrast, carbamazepine administration results in more stable networks that do not return to resting baseline. The use of more controllable MPS paradigms - ideally allowing for prolonged testing times - will be required to interpret these results, and forms an important direction for future research. Providing mechanistic interpretations of nuanced changes observed in these dynamic systems

exceeds the scope of this specific research. Despite this limitation, these results do showcase the utility of conducting pharmacological assays in neural cultures when assessing SBI characteristics in a closed-looped structured information landscape. While the preclinical relevance of the metrics, such as functional connectivity, is yet to be conclusively determined, the additional data provided by an SBI approach can allow future

Fig. 7 | Network summary statistics using all of the recorded channels under control and different drug treatments. Comparing gameplay and rest recordings: **a–d** average weight, **e–h** modularity Index, and **i–l** clustering coefficient values for gameplay and rest recordings under control and all pharmacological interventions using the entire recording length from all 1024 recorded channels. One-way ANOVA test within each group and post-hoc Tukey's test between all the rest and all the full game groups. # or * indicate $p < 0.05$, # or ** indicate $p < 0.01$, ### or *** indicate $p < 0.001$. Network summary statistics between the first and last 5 minutes of full game and rest recordings using all of the recorded channels under different

drug treatments. Comparing the first and last 5 minutes of full game: **m–o** Average weight, **p–r** Modularity Index, and **s–u** Clustering coefficient values for the first and last 5 minutes of recordings under all pharmacological intervention using all 1024 recorded channels. One-way ANOVA test within each group and post-hoc Tukey's test between all the rest and all the full game groups. # or * indicate $p < 0.05$, ## or ** indicate $p < 0.01$, ### or *** indicate $p < 0.001$. Boxplots show interquartile range, with bars demonstrating $1.5\times$ interquartile range, the line marks the median, and the black triangle marks the mean. Error bands = 1 SE.

research to create more predictive models to better understand how neural cultures are influenced by pharmacological intervention.

Apart from the specific limitations already discussed, this paper also acknowledges several notable general constraints. Foremost, the in vitro neural culture model adopted for this paper is a 2-dimensional monolayer of predominantly a single neural cell type. While this allowed rapid generation and iterations in testing different pharmacological compounds at a reasonably high throughput, it is likely that this does not capture the more complex activity that occurs in epilepsy. The generation of 3-dimensional neural organoids in simulated information-rich environments has previously been proposed as a method called Organoid Intelligence (OI)^{38–40}. An OI approach may offer the ability to determine even more complex interactions between the neural systems and pharmacological interventions. Additionally, future work could also consider generating and evaluating in vitro models from donors who display genetically related forms of epilepsy^{19,41,42}. Another key limitation of this work is the limited time window during which testing could be undertaken; as previously explored, the *DishBrain* setup allows for restricted testing periods as the heat generated by the multielectrode array (MEA) system results in evaporation, changes in osmolarity of the cell culture media, and potentially electrolysis at the stimulating electrode surface⁴³. Therefore prolonged testing exceeding that used in this study eventually result in degradation of cell health and eventual cascades of cell death in the cultures. Future work should aim to use systems that either generate minimal heat or adopt an approach of embedding neural cultures in closed-system perfusion circuits capable of constant homeostatic control of the cell culturing environment to allow for longer-term testing of cultures. Further, while the drugs chosen for this study represent established options for the treatment of epilepsy and have been used in similar preclinical drug screens involving NGN2 iPSC-derived neurons²⁰, a more comprehensive panel of ASMs can be identified, including those with differing mechanisms of action. As such, future work could utilize wider libraries of compounds to gain a deeper insight into the effects of suppression of neuronal excitability on this system. Finally, this work intends to use the previously validated *DishBrain* system to assess the information processing capabilities of a neural system that expresses glutamatergic hyperactivity in the context of epilepsy. While improvements in the cell culture model remain possible for future work, the differences in neural activity between gameplay and rest conditions and the impact of various pharmacological interventions were demonstrated across multiple metrics. Of course, it is possible that using enhanced disease models, alternative pharmaceutical interventions, or different stimulation and recording settings might yield different performance outcomes. As such, future work should aim to use more modifiable microphysiological systems capable of trialing different learning environments and rapidly iterating through different stimulation and recording assays.

Ultimately, this study demonstrated that pharmacological modulation of a human iPSC-derived model of glutamatergic hyperexcitability can influence both electrophysiological activity and information processing within a structured, closed-loop environment. By integrating real-time stimulation and feedback through the *DishBrain* system, we captured dynamic responses to ASMs that traditional in vitro assays cannot easily resolve. Despite some technical and biological limitations, our results underscore the value of an SBI approach in preclinical research. This method not only enables richer functional assessments of drug effects but also lays the groundwork for more predictive, human-relevant models of neurological

disease. As MPS platforms evolve, combining improved cellular models, extended recording times, and adaptive closed-loop paradigms may transform how we evaluate therapeutic interventions, bridging the gap between neuropharmacological modulation and meaningful improvements in information processing.

Methods

iPSC culture

All cultures were maintained in a 37 °C, 5% CO₂, 5% O₂ humidified cell culture incubator (BINDER, Germany). NGN2 iPSCs were thawed on vitronectin (VTN, ThermoFisher, Australia)-coated T25 flasks consisting of 18 µL VTN in 1.8 mL Dulbecco's phosphate buffered saline, DPBS, containing Ca²⁺/Mg²⁺ and washed with 4 mL DPBS without Ca²⁺/Mg²⁺ (DPBS–/–), prior to cell seeding and maintained in 4 mL StemFlex iPSC media (ThermoFisher, Australia). Cells were passaged using EDTA (0.5 mM in DPBS–/–). NGN2 hiPSCs were generously provided by Mark Kotter, and all experiments were performed in accordance with relevant guidelines and regulations related to the use of hiPSCs.

Neuronal differentiation

To initiate differentiation, NGN2 iPSCs²² were dissociated with Accutase (Merck, Australia) and seeded on Geltrex (ThermoFisher, Australia)-coated 24-well plates at a concentration of 80,000 cells per cm² in 1 mL StemFlex media with 5 µM Y-27632. The next day (day 0), cells were washed with 1 mL DPBS –/– and media changed to Day 0–1 media (Table 1) containing doxycycline 2 µg/mL (DOX, Merck, Australia). Cells were maintained in this media until day 2, when cells were washed with 1 mL DPBS –/– and media changed to Day ≥2 media (Table 1) containing DOX 2 µg/mL. On day 3, cells were washed with DPBS –/–, dissociated with Accutase and plated onto Poly-D-Lysine-Geltrex coated MaxONE multielectrode array chips (Maxwell BioSystems, Switzerland) in Day ≥2 media (Table 1) containing DOX (2 µg/mL) and 5 µM Y-27632. 20000 NGN2 cells, and 50000 primary human cortical astrocytes (Sciencell Cat# 1800, USA; maintained previously according to manufacturer's specifications) were mixed and plated on each MEA chip at this time. On days 4 and 5, full media changes in Day ≥2 media + DOX were performed. On day 7, media was changed to Day ≥2 media without DOX. Thereafter, half media changes were performed every second day.

Immunocytochemistry

From day 35 onwards, cultures grown in parallel on plastic tissue culture-treated 24-well plates were washed free of media (3 × 5 min in DPBS –/–) and fixed in 4% paraformaldehyde aqueous solution (Electron Microscopy Services, USA) for 15 min at room temperature (RT; 20 °C). Wells were washed once with DPBS–/– and then permeabilized with 0.1% Triton X-100 (Merck, Australia) in DPBS–/– for 30 min at RT. Cells were then blocked with 1% normal goat serum in DPBS–/– for 30 min and incubated overnight at 4 °C in 0.1% Triton X-100 in DPBS–/– with the primary antibodies shown in Table 2. The next day, after another 3 washes, secondary antibodies as shown in Table 2 were applied for 2 h at room temperature. Cells were again washed three times and the nucleic acid stain DAPI (ThermoFisher, Australia) was briefly added for 5 min prior to visualization to indicate the presence of cells. Cultures were visualized using a Nikon A1Plus Ti ZDrive confocal microscope (Nikon, Japan), and at least

Table 1 | Cell culture media used in neural differentiation in the present study

Formula	Item	Amount
Day 0–1 media	DMEM/F12 (11330057, ThermoFisher, Australia)	50 mL
	P/S (15140–163, ThermoFisher, Australia)	500µL
	NEAA (11-140–050, ThermoFisher, Australia)	500 µL
	Glutamax (35050038, ThermoFisher, Australia)	500 µL
	2-ME (31350010, ThermoFisher, Australia)	50 µL
	N2 supplement (17502-048, ThermoFisher, Australia)	500 µL
	Doxycycline (2 mg/mL, D9891, Merck, Australia)	50 µL (2 µg/mL)
Day ≥ 2 media	Neurobasal (21103049, ThermoFisher, Australia)	50 mL
	P/S (15140–163, ThermoFisher, Australia)	500 µl
	Glutamax (35050038, ThermoFisher, Australia)	500 µl
	2-ME (31350010, ThermoFisher, Australia)	50 µl
	B27 supplement (17504044, ThermoFisher, Australia)	1 mL
	BDNF (50 µg/mL, 78005, STEMCELL Technologies, Canada)	10 µl (10 ng/ml)
	NT3 (50 µg/mL, 78074, STEMCELL Technologies, Canada)	10 µl (10 ng/ml)
	Doxycycline (2 mg/mL, D9891, Merck, Australia)	50 µL (2 µg/mL)

Table 2 | Antibodies used in immunocytochemistry experiments

Group	Antibody	Usage
Primary	Anti-VGLUT2 (ab79157, Abcam, USA)	1:1000
	Anti-FOXG1 (ab196868, Abcam, USA)	1:100
	Anti-MAP2 (ab5392, Abcam, USA)	1:5000
	Anti-PSD-95 (ab192797, Abcam, USA)	1:1000
	Anti-SYN1 (ab254349, Abcam, USA)	1:500
	Anti-GFAP (ab4674, Abcam, USA)	1:500
	Anti-TUJ1 (MAB1637, Merck, Australia)	1:1000
Secondary	Gt anti rabbit 488 (ab150077, Abcam, USA)	1:1000
	Gt anti chicken 555 (ab159170, Abcam, USA)	1:1000
	Gt anti mouse 647 (SAB4600183, Merck, Australia)	1:1000
Stain	DAPI 1 mg/mL (62248, ThermoFisher, Australia)	1:10000

three images (2048 x 2048 pixels with 161 z-steps of 0.7 µm) were recorded per well.

Single-cell RNA sequencing

Human iPSCs (D0) and iNeurons (days 7, 14, and 21) were washed once with 1× DPBS –/– before adding either Accutase (day 0) or Accutase containing 20 units/mL of papain, 5 mM MgCl₂ and 5 mg/mL DNase I (Worthington, United Kingdom) (days 7–21). The cells were incubated at 37 °C for up to 5 min (day 0) or up to 45 min (days 7–21) before adding DMEM/F12 (ThermoFisher, United Kingdom) supplemented with 10% fetal bovine serum (Merck, United Kingdom). The cells were dissociated using a P1000 and collected in a 15 mL tube. After centrifugation, the cells were resuspended in 1× DPBS –/– containing 1% BSA (Merck, United Kingdom) and 10 µM ROCK inhibitor (Y-27632, Tocris, United Kingdom) and collected in a 15 mL tube capped with a 40 µm cell strainer. Following centrifugation, cells were washed three additional times. Single cell suspensions were counted using an automated cell counter (Countess, Invitrogen, United Kingdom) and concentrations adjusted to 1000 cells/mL. Cells were then hash-tagged using total-seq anti-human hashtag antibody (BioLegend, United Kingdom) and kept on ice for further processing.

Single-cell suspensions were processed by the Chromium Controller using Chromium Single Cell 3’ Reagent Kit v2 by 10x Genomics. Each

sample was loaded for a target recovery of 8000 cells per sample. All the steps were performed according to the manufacturer’s specifications. All samples were sequenced using the Illumina HiSeq platform (Illumina, USA), with 50,000 reads on average per cell. scRNAseq samples were processed using the 10x Genomics CellRanger pipeline v3.0.1 using the GRCh38 genome assembly to produce the gene expression matrices. The filtered gene expression matrix was used without modifying any cell gating parameters.

Standard quality control was performed in R (v.4.3.1) using Seurat (v.4.4.0): Cells with mitochondrial RNA below 5% and with 200 to 10000 genes were included. Counts were scaled, and mean expression levels at each time point were normalized to day 0 (hiPSCs) for visualization. All data used in the analysis can be accessed at the Open Science Framework (OSF) repository: <https://osf.io/xwf7v>⁴⁴.

Electrophysiology and DishBrain pharmacology

MEA chip activity was assessed by performing activity recordings using the MaxWell MaxONE system and software (Maxwell BioSystems, Switzerland) at regular timepoints during neuronal differentiation. Briefly, MEA chips were removed from the cell culture incubator and placed on one of three MaxONE readers in a non-humidified 37 °C 5% CO₂ incubator and equilibrated for 10 min before 5 min of full recording. As reported previously⁷ when chips obtained a culture-wide average activity of 0.7 Hz (generally between day 21 and day 35 of differentiation), chips were considered ready for DishBrain gameplay. On days where gameplay was performed, cells were removed from the cell culture incubator, and an activity assay was conducted as above to ensure the minimum required activity for gameplay was present. If the chip passed, cells were treated with one of the following compounds (all using DMSO as a solvent, and diluted such that final DMSO concentration did not exceed 0.1% (v/v)): phenytoin 10–33 µM, perampanel 1–10 µM, carbamazepine 2–200 µM, before being returned to the cell culture incubator. For vehicle control chips, DMSO was added at a final concentration of 0.1% (v/v). After 1 h, cells were returned to the reader for another activity assay, then closed-loop feedback gameplay was started as follows: Rest session (15 min), Active session (15 min), Rest session (15 min), Active session (15 min). Afterwards, chips had a full media change (1 mL) and were returned to the cell culture incubator. In Rest sessions, cell responses were recorded and contributed to gameplay, but cells were not stimulated with information encoding the gameplay world. In Active sessions, this information was also provided to the cells. (When referring to “gameplay” generally, we are talking about the DishBrain assay itself, and in the context of gameplay results, we are describing the culture’s

performance in the Active session with respect to the Rest session, unless specifically noted in the text or figure. Further details on stimulation and configuration parameters during Rest and Active gameplay are presented in Supplementary Methods.) Chips were exposed to 60 min gameplay conditions a maximum of one time per day, and played for up to four consecutive days.

Statistics and reproducibility

Activity assay recordings were analyzed to measure average spike firing rate and average spike amplitude. From gameplay experiments, hit/miss ratio and rally length were calculated, and results were summarized for each treatment (drug/dose) group. Summary data is presented as the mean hit/miss ratio or rally length, the standard error of the mean, and the count (n) of the treatment observations. Significance testing was performed to determine differences, if any, between treatment groups (Student's t test or one-way ANOVA with post-hoc Bonferroni multiple comparison test compared to DMSO vehicle control). A p value <0.05 was taken as the minimum level of significance. Replicate number counts (n) generally refer to the sample size of independent gameplay experiments performed, unless otherwise noted in the figure caption. The majority of data analysis was performed in Python as described in the section "Results". Initial electrophysiological analyses were performed using the RStudio (Posit Software, PBC) Build 463 IDE with R version 4.3.1 and the following packages: tidyverse version 2.0.0, ggsignif version 0.6.4. The statistics tables provided in the Supplementary Data detail all comparisons, standard errors, degrees of freedom, p values and other related inferential statistical values.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The imaging data supporting this study are presented in the figures; original high-resolution versions are available from the corresponding author upon reasonable request. RNA-Seq data files and the electrophysiological datasets generated and analyzed during this study are available at the OSF repository: <https://osf.io/xwf7v>⁴⁴. The statistics tables provided in the Supplementary Data detail all comparisons, standard errors, degrees of freedom, p values and other related inferential statistical values.

Code availability

The code supporting the analysis and figure generation in this study is publicly available at <https://osf.io/xwf7v>. Code related to the proprietary DishBrain platform is not publicly available due to intellectual property restrictions. The high-level structure of DishBrain can be found in the Supplementary Methods and in Supplementary Fig. 1.

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Author contributions

B.W. and B.J.K. conceived the project. B.W., F.H., K.B., M.K., and B.J.K. designed the experiments. K.B. and M.K. generated and provided the NGN2

iPSC line. B.W. and C.D. performed the *DishBrain* experiments. N.P. and K.B. performed the RNA-Seq experiments. B.W., F.H., N.P., M.Kh., K.B., and B.J.K. analyzed the data. F.H. and M.Kh. conducted the neurocomputational analysis. B.W., F.H., C.D., M.Kh., A.L., K.B., N.P., M.K., and B.J.K. contributed to writing the manuscript.

Competing interests

The authors declare the following competing interests: B.W., F.H., C.D., A.L., and B.J.K. are employed by Cortical Labs Pte Ltd, a for-profit company interested in the commercial viability of synthetic biological intelligence and related patents. B.W., F.H., A.L., and B.J.K. also hold a pecuniary interest in Cortical Labs Pte Ltd. MRK owns shares in bit.bio and in Cortical Labs and is a director of bit.bio Ltd.

Additional information

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Correspondence and requests for materials should be addressed to Brett J. Kagan.

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